

Highlights

- Six HDPE/high-Z nanocomposites were benchmarked in Geant4.
- Effective-medium models matched XCOM within 0.339-0.793% mean error.
- Parameterised 1 μm RVEs passed the 3% H1 gate for five binaries.
- Multi-union RVEs were useful only as small finite-RVE checks.
- A ternary negative control showed why composition equivalence matters.